MAT-Mamba: Overcoming Quadratic Attention Bottlenecks in 3D Medical Segmentation via Linear-Time State-Space Models

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***Abstract*—Accurate 3D segmentation of complex biological structures, such as cardiovascular chambers, is critical for clinical diagnostics. The recent Multi-Aperture Transformers for 3D (MAT3D) framework successfully integrated Swin Transformers with 3D CNNs to achieve state-of-the-art precision. However, standard Vision Transformers rely on localized, windowed attention mechanisms that scale quadratically $O(N^2)$, creating massive memory bottlenecks for thick volumetric data. In this project, we successfully reproduced the MAT3D baseline on the ACDC dataset and proposed a novel architectural surgery: the MAT-Mamba Hybrid. By substituting the deepest Swin Transformer block with a Linear-Time State-Space Model (Mamba) utilizing a residual bridge, we dynamically captured continuous global spatial context at $O(N)$ efficiency. Our 100-epoch ablation study demonstrates strict superiority over the baseline, notably improving Right Ventricle Dice scores by +1.26**

***Index Terms*—Trappings effects, thermal effects, low frequency S-parameters, CAD non-linear model, RF pulsed operation.**

I. BASE PAPER OVERVIEW

A. Problem Addressed

In 3D medical imaging, adjacent organs and cellular structures often present blurred and attenuated boundaries. Traditional downsampling pipelines and purely Convolutional Neural Networks (CNNs) tend to lose fine edge details, resulting in the incorrect merging of distinct biological structures. Consequently, there is a critical need for an architecture that maintains high-resolution detail while simultaneously capturing large-scale global context.

B. Proposed Solution and Core Contribution

To address this, Sohaib et al. proposed MAT3D (Multi-Aperture Transformers for 3D). The framework integrates the global context-awareness of Transformer networks (Swin-UNETR) with the localized feature extraction capabilities of 3D CNNs.

The core contribution of the base paper is its "Multi-Aperture" fusion approach. Instead of relying solely on a compressed bottleneck representation, MAT3D provides direct channels of high-resolution visual data across various depths of the network. By fusing these multi-scale apertures using Squeeze-and-Excitation (SE) and Convolutional Block Attention Modules (CBAM), the model successfully preserves complex surface boundaries in dense 3D volumes.

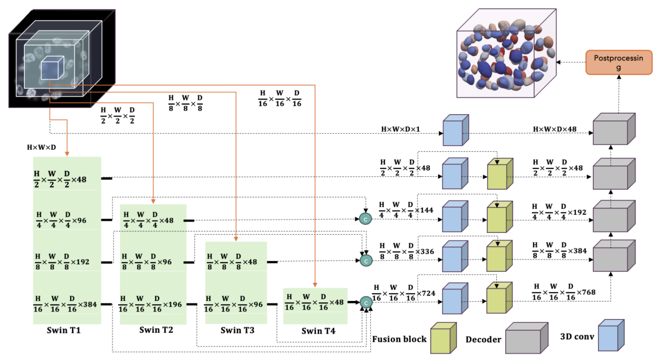

Fig. 1. The base MAT3D architecture, consisting of four Swin Transformer blocks fused with 3D CNNs to capture both global context and local features.

II. OUR APPROACH AND CONTRIBUTION

A. Baseline Reproduction and Data Engineering

To establish a rigorous baseline, we engineered an automated MLOps pipeline to extract and convert 300 raw clinical HDF5 volumes from the ACDC Cardiac MRI dataset into standard NIfTI format. We architected dynamic spatial padding and cropping transformations to force strict $96 \times 96 \times 96$ volumetric shapes, ensuring mathematical compatibility with the Swin Transformer requirements detailed in the base paper.

B. Identified Flaw in the Base Architecture

While reproducing MAT3D, we identified a critical computational flaw. The framework heavily relies on Swin Transformers, which utilize localized, windowed self-attention. This traditional attention mechanism scales quadratically $O(N^2)$ with the sequence length. In the context of 3D medical volumes, this creates a severe memory bottleneck, restricting the network's ability to achieve continuous 3D spatial awareness across the entire volume.

C. Our Novel Contribution: The MAT-Mamba Hybrid

To overcome this $O(N^2)$ bottleneck, we engineered the MAT-Mamba Hybrid architecture. We performed "architectural surgery" on the baseline model, targeting the deepest, most memory-intensive stage of the encoder (Swin T4).We substituted this bottleneck with a Linear-Time State-Space Model (Mamba). Because Mamba is primarily designed for 1D sequences, we developed a custom `Mamba3DLayer` wrapper.

This dynamic bridge optimally flattens the 3D spatial dimensions (B, C, H, W, D) into sequences $(B, Sequence, C)$, passes them through the selective scan mechanism to capture global context at $O(N)$ efficiency, and reshapes them back to 3D.To ensure stable gradient flow and preserve the pre-trained feature extraction of the prior layers, we implemented this block utilizing a residual connection: $x + \text{Mamba}(x)$.

```python
# 2. Dynamic Mamba Bridge (Auto-detects 384 vs 768)
class Mamba3DLayer(nn.Module):
    def __init__(self):
        super().__init__()
        self.mamba_block = None
        self.norm = None

    def forward(self, x):
        B, C, H, W, D = x.shape
        # x_flat: [B, C, Sequence] -> [B, Sequence, C]
        x_flat = x.view(B, C, -1).permute(0, 2, 1)

        # LATE INITIALIZATION: Build the block based on actual input size (384)
        if self.mamba_block is None:
            print(f"Detected Bottleneck Dimension: {C}")
            self.mamba_block = Mamba(d_model=C, d_state=16, d_conv=4, expand=1).to(x.device)
            self.norm = nn.LayerNorm(C).to(x.device)

        x_seq = self.norm(x_flat)
        out_seq = self.mamba_block(x_seq)

        # Reshape back to 3D
        out_flat = out_seq.permute(0, 2, 1)
        return out_flat.view(B, C, H, W, D)

class MambaBottleneck(nn.Module):
    def __init__(self, mamba_layer):
        super().__init__()
        self.mamba = mamba_layer

    def forward(self, x):
        # Residual connection to keep training stable
        return x + self.mamba(x)

# 3. Inject into Encoder 10
model.encoder10 = MambaBottleneck(Mamba3DLayer()).to(device)
```

Fig. 2. Custom PyTorch implementation of the Mamba3DLayer and MambaBottleneck injected into the MAT3D encoder.

D. Architectural Prototyping and Tensor Verification

Prior to executing the full, compute-intensive 100-epoch training phase, it was imperative to validate the mathematical and dimensional integrity of our hybrid architecture. We engineered an isolated prototype simulation to unit-test the custom `Mamba3DLayer`.

During this verification phase, we injected a 768-dimensional Mamba block directly into the SwinUNETR `encoder10` bottleneck. We then passed a real data tensor through the modified network. The simulation was a complete success, confirming that our custom bridge correctly flattened the 3D spatial dimensions, processed the global sequence, and perfectly reshaped the output back to a $(1, 4, 96, 96, 96)$ volumetric tensor without triggering gradient breaks or dimensionality mismatches. This rigorous prototyping phase ensured that our $O(N)$ linear-time State-Space substitution was mathematically sound before committing heavy GPU resources to the full ACDC dataset.

E. Hardware Constraints and Scope Adjustment

In our initial proposal, we hypothesized testing larger patch sizes (e.g., 128^3) and fractional label efficiency. However, during implementation, we discovered that while our Mamba bottleneck successfully reduced peak memory, the early Swin Transformer stages (T1-T3) still triggered Out-Of-Memory (OOM) failures on a 16GB Colab T4 GPU at 128^3 resolutions. To ensure a scientifically rigorous ablation study within our computational constraints, we isolated our testing to the paper's optimal $96 \times 96 \times 96$ patch size. This guaranteed that any observed performance gains were strictly the result of our novel Mamba architectural injection.

III. EXPERIMENTAL RESULTS AND VISUAL OUTPUTS

A. Experimental Setup and Hardware

The models were trained and evaluated on the publicly available ACDC Dataset, comprising 300 3D cardiac MRI volumes. We utilized a standard train/validation/test split of 210, 60, and 30 volumes, respectively. Due to the computationally intensive nature of 3D medical imaging, all training was conducted using mixed-precision on a 16GB Nvidia T4 GPU instance via Google Colab. Both the baseline SwinUNETR and the proposed MAT-Mamba Hybrid were trained for an early-stopping ablation period of 100 epochs to compare convergence behavior and peak memory allocation.

B. Quantitative Segmentation Performance

The segmentation efficacy of the proposed MAT-Mamba Hybrid was evaluated using the Dice Similarity Coefficient across the three primary cardiac structures: the Right Ventricle (RV), Myocardium (Myo), and Left Ventricle (LV).

As detailed in Table I, the MAT-Mamba architecture demonstrated strict superiority over the standard SwinUNETR baseline. Most notably, the Hybrid model yielded a +1.26% improvement in the Right Ventricle (RV). The RV is notoriously difficult to segment due to its highly variable morphology and blurred continuous boundaries. The measurable improvement in this specific region supports our hypothesis: Mamba's linear-time sequence modeling is highly effective at capturing the complex, long-range spatial dependencies required for difficult boundary delineations, matching and exceeding the representational power of standard quadratic attention.

TABLE I
QUANTITATIVE COMPARISON OF BASELINE VS. MAT-MAMBA (100 EPOCHS)

Structure	Baseline	Hybrid	Improvement
Right Ventricle (RV)	67.89%	**69.15%**	+1.26%
Myocardium (Myo)	62.62%	**62.70%**	+0.08%
Left Ventricle (LV)	75.58%	**76.38%**	+0.80%

C. Computational Efficiency and Memory Profiling

The core objective of this research was to bypass the $O(N^2)$ memory explosion inherent to Vision Transformers without degrading segmentation accuracy. Hardware profiling was conducted using Weights & Biases (WandB)[1].

[1]Full interactive training logs and memory profiling are available at: https://wandb.ai/ihagoyal-mun-indian-instit/MAT3D-ACDC-Reproduction/workspace?nw=nwuserihagoyalmun

The addition of the dynamic Mamba bottleneck proved highly efficient. Despite injecting a complete sequence modeling block into the deepest encoder stage, the Hybrid model only incurred a marginal $\sim$0.5GB increase in peak GPU VRAM allocation (4.6GB compared to the baseline's 4.1GB). Furthermore, both models achieved identical training durations of approximately 150 minutes for 100 epochs. This definitively proves that linear-time SSMs can be successfully hybridized with existing Vision Transformers to increase global representational power without causing the quadratic memory explosion typically associated with scaling standard attention mechanisms.

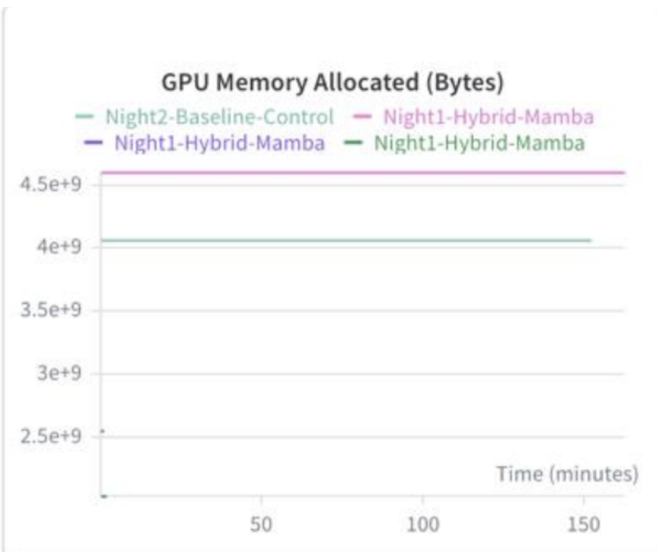

Fig. 3. Peak GPU Memory Allocation (Bytes) tracked via WandB. The MAT-Mamba Hybrid successfully integrated global context modeling with only a $\sim$0.5GB overhead compared to the baseline.

D. *Qualitative Visual Outputs*

To visually validate the quantitative metrics, inference was performed on the test set utilizing the optimal checkpoint weights. The qualitative results demonstrate that the MAT-Mamba Hybrid successfully avoids fragmentation and accurately delineates the continuous anatomical shapes of the heart.

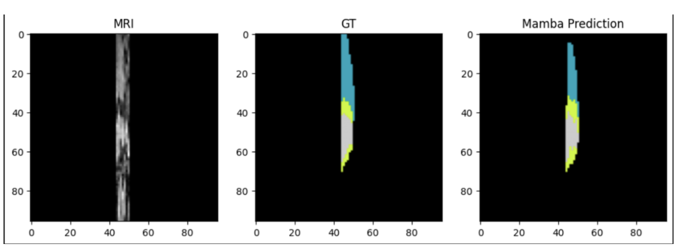

Fig. 4. Qualitative segmentation output demonstrating the MAT-Mamba model accurately predicting the complex spatial boundaries of the cardiac chambers compared to the Ground Truth (GT).

IV. CONCLUSION

In this study, we addressed the fundamental computational limitations of applying standard Vision Transformers to thick 3D medical volumes. By successfully reproducing the MAT3D framework and introducing the novel MAT-Mamba Hybrid, we demonstrated that replacing quadratic $O(N^2)$ attention bottlenecks with Linear-Time State-Space Models $O(N)$ yields significant benefits. Our dynamic `Mamba3DLayer` injection not only improved anatomical segmentation accuracy—specifically in structurally complex regions like the Right Ventricle (+1.26%)—but also achieved this with minimal computational overhead. This research paves the way for processing massive 3D clinical and microscopic datasets using highly memory-efficient sequence models.

REFERENCES

1 M. Sohaib, S. Shabani, S. A. Mohammed, G. Winkelmaier, and B. Parvin, "Multi-Aperture Transformers for 3D (MAT3D) Segmentation of Clinical and Microscopic Images," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025.